Multi-level Context-aware Reasoning for enhancing Visual Emotion Analysis: Decoding Image-Inherent Sentiment-Stimuli

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Abstract

Visual Emotion Analysis (VEA) aims to perceive the emotion of people towards an image, which becomes important with the development of social media. Most existing methods regard it as a classification task and rely on the manual labels for supervision. We consider that visual emotion involves a highly sophisticated and abstract cognitive process; simple classification leads to sub-optimal results due to the affective gap. Inspired by the recent psychological research which discovered a set of visual emotional cues, we propose a Multi-level Context-aware Reasoning Stimuli framework to comprehensively decode the sentiment signals, which is referred to as MCRStimuli. Specifically, we stratify image-inherent sentiment-stimuli into a three-tier hierarchy: pixel-level, semantic-level, and cognitive-level, which correspond to different contextual information and reflect successive stages of visual abstraction and affective elaboration. To this end, we leverage a Vision-Language Model (VLM) to generate the corresponding emotional features for different-level stimuli, ensuring each tier contributes disentangled yet complementary evidence for enhancing VEA. Compared with the recent state-of-the-art methods, which rely on class-conditional prompts or visual captions, our MCRStimuli achieves significant improvements on multiple benchmarks, which demonstrate the importance of decoding the reliable sentiment-stimuli. We also demonstrate its superiority in generalizing to different visual domains, providing valuable insights into robust visual emotion understanding.

1 Introduction

With the prevalence of social media, people's emotions can be conveyed within the images [Han *et al.*, 2022; Xie *et al.*, 2024]. Therefore, visual emotion analysis (VEA) receives great attention, which aims to predict the dominant emotions triggered by an image. It is essential in endowing machines with the ability to understand emotions. Therefore, developing a robust VEA framework is vital for various downstream tasks of human-machine interactions, *e.g.,* emotional advertising [Shukla *et al.*, 2022], mental health analysis [Fei *et al.*, 2020], social robotics [Wang *et al.*, 2023a].

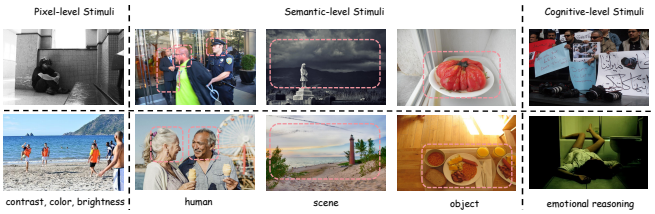

Figure 1: Illustration of visual examples which are associated with different levels of stimuli. (see text for details)

Unlike conventional computer vision tasks, which focus on recognizing the objective information, VEA seeks to capture the subjective emotional information, which is more abstract and challenging [Yang *et al.*, 2018; Lee *et al.*, 2025]. Most previous methods [Zhang *et al.*, 2019; Yang *et al.*, 2021a] regard it as a classification task, which aims to automatically extract sentiment-related visual features under the supervision of manual labels. However, there exists a significant affective gap [Zhao *et al.*, 2014b] between the pixel objects and the inherent emotions, thus it is difficult to establish the correct relationship between the emotion and the traditional visual features [Xu *et al.*, 2022; Wang *et al.*, 2023b].

Therefore, it would be essential to decode the image-inherent stimuli that evoke emotions [Yang *et al.*, 2021a; Yang *et al.*, 2023] for more reliable visual emotion analysis. According to previous psychological studies [Ritchie and Batteson, 2013; Kurt *et al.*, 2017], the emotions are first evoked by the global pixel-level stimuli, *e.g.,* colorfulness and brightness (Figure 1(left)). Second, when there exist scenes, objects, and humans, the psychological researchers [Brosch *et al.*, 2010; Frijda, 2009] suggest that the interactions among them play a key role in emotional stimulation (Figure 1(middle)). Last, [Lindquist *et al.*, 2015a; Lindquist *et al.*, 2015b] discovered that human emotion is intertwined with human language, which serves the cognitive and reasoning process, and links abstract affective information into accessible emotion concepts [Deng *et al.*, 2024; Lee *et al.*, 2025; Zhou *et al.*, 2025b]. Therefore, for complex images (Figure 1(right)) where visual cues are ambiguous,

cognitive reasoning that interprets contextual events (people protesting and unnatural body postures) is essential for arriving at the correct emotional predictions (anger and fear).

Based on the above studies, we propose a Multi-level Context-aware Reasoning Stimuli framework (MCRStimuli), which aims at comprehensively decoding the image-inherent sentiment-stimuli from different levels (pixel-level, semantic-level, and cognitive-level). The three-tiered sentiment-stimuli reflect successive stages of visual abstraction and affective elaboration, and correspond to different contexts, as shown in Figure 1. In addition, different from recent methods [Yang *et al.*, 2021a; Yang *et al.*, 2021b] which used the pre-trained vision models for extracting emotional cues, we introduce a novel self-generation pipeline to reason the stimuli from different levels by leveraging a large Vision-Language Model (VLM) [Wang *et al.*, 2024]. Specifically, given an image, we first enforce the VLM to perceive the colorfulness, brightness, and contrast information as the pixel-level stimuli. Then, we focus on selecting the main scenes, objects, and humans and understanding their interactions at the semantic-level stimuli. Finally, these surficial objective sentiment stimuli are used for linguistic emotional reasoning, which is regarded as the cognitive-level stimuli. We further perform a fusion operation, allowing multi-level stimuli to contribute in a disentangled yet complementary manner and substantially enhancing emotion understanding. In this paper, instead of predicting emotions directly from visual features, MCRStimuli streamlines the process of VEA, enabling more effective and interpretable emotion prediction. We demonstrate that our framework can consistently outperform the recent state-of-the-art methods on multiple visual emotional benchmarks, and is more robust when testing on different visual domains. To summarize, the contributions are listed as follows:

1. We propose a novel MCRStimuli framework, which comprehensively decodes the image-inherent sentiment-stimuli and achieves state-of-the-art performances.
2. We design a self-generation pipeline with a VLM based on the unique properties of VEA, and the reasoned stimuli improve the robustness by emotion understanding.
3. These multi-level stimuli, serving as the missing attributes for corresponding visual datasets, will be made public. They are intended to foster further community research by providing essential insights. https://github.com/Raymond-Mak/-MCRStimul_captions

2 Related Work

2.1 Generic Visual Emotion Analysis

Previous methods on VEA mainly focused on learning affect-discriminative representations, which include designing hand-crafted features [Machajdik and Hanbury, 2010; Borth *et al.*, 2013] or using deep learning techniques to extract sentiment-related features, automatically [Zhang *et al.*, 2019; She *et al.*, 2019; Yang *et al.*, 2021a]. For example, [Borth *et al.*, 2013] proposed the SentiBank to figure out emotional visual concepts with Adjective Noun Pairs (ANPs). [Zhao *et al.*, 2014a] proposed to extract emotional features based on low-level artistic principles. Despite their improvements, these hand-crafted features are insufficient to guarantee reliable performance due to the complexity of visual emotion. In addition, [Chen *et al.*, 2014] extended the SentiBank with a deep learning model for training the visual sentiment concept classification. [Yang *et al.*, 2021a; She *et al.*, 2019] proposed to extract salient entities and regions for fine-grained affect inference. Inspired by the Stimuli-Organism-Response (S-O-R) emotion model, [Yang *et al.*, 2021b] leveraged three specific networks to extract color, object, and face features for the emotion prediction. [Xu *et al.*, 2022] incorporates hierarchical emotion knowledge to mitigate the affective gap.

2.2 VLM-assisted Visual Emotion Analysis

With the rise of visual-language models (e.g., Qwen-VL [Wang *et al.*, 2024], BLIP2 [Li *et al.*, 2023], LLaVA [Liu *et al.*, 2023]), which demonstrate impressive performance on visual understanding, fully exploiting these models in VEA becomes a new and interesting direction due to the close relationships between emotion and language. For example, [Deng *et al.*, 2023; Deng *et al.*, 2024] proposed to use a class-specific text prompt as additional supervision signals with CLIP [Radford *et al.*, 2021; Bustos *et al.*, 2023]. [Cen *et al.*, 2024] designed a Multi-Aspect Semantic Auxiliary Network which fuses the features from the image and the multi-aspect text descriptions. Multi-Views Prompt Learning (MVP) [Xu *et al.*, 2024] shared a similar idea by incorporating a learnable multi-view contextual prompt for enhancement. Meanwhile, [Xie *et al.*, 2024] and [Lee *et al.*, 2025] proposed to leverage visual instruction tuning with self-generated emotion instruction data, which demonstrated proficiency in emotion classification. [Zhou *et al.*, 2025b; Etesam *et al.*, 2024] proposed to generate detailed captions to augment visual understanding and emotion prediction. More recently, [Zhou *et al.*, 2025a] proposed to make emotion prediction based on pure texts from the OCR model or the emotional attribute chain-of-thought prompting. In contrast, our method pioneers the use of VLM to reason multi-level sentiment-stimuli according to the psychological studies, which is more comprehensive than the generic image captions. We demonstrate that these sentiment-stimuli are crucial for better visual emotion understanding, thus yielding state-of-the-art performance and allowing the model to robustly adapt to different domains.

3 Method

Given an image x, our goal is to infer the emotion label y. Recent works treated it as a visual classification task and mainly relied on the visual features with the help of the objector detector [Yang *et al.*, 2021a] or face expression network [Yang *et al.*, 2021b], etc. We further consider that incorporating additional language descriptions, which is more associated with human emotion [Lindquist *et al.*, 2015a; Lindquist *et al.*, 2015b], for better visual emotion understanding. However, the general objective- or class-level text descriptions are limited in covering all important stimuli in visual emotions. Therefore, decoding the image-inherent sentiment-stimuli is essential for enhancing the performance of VEA.

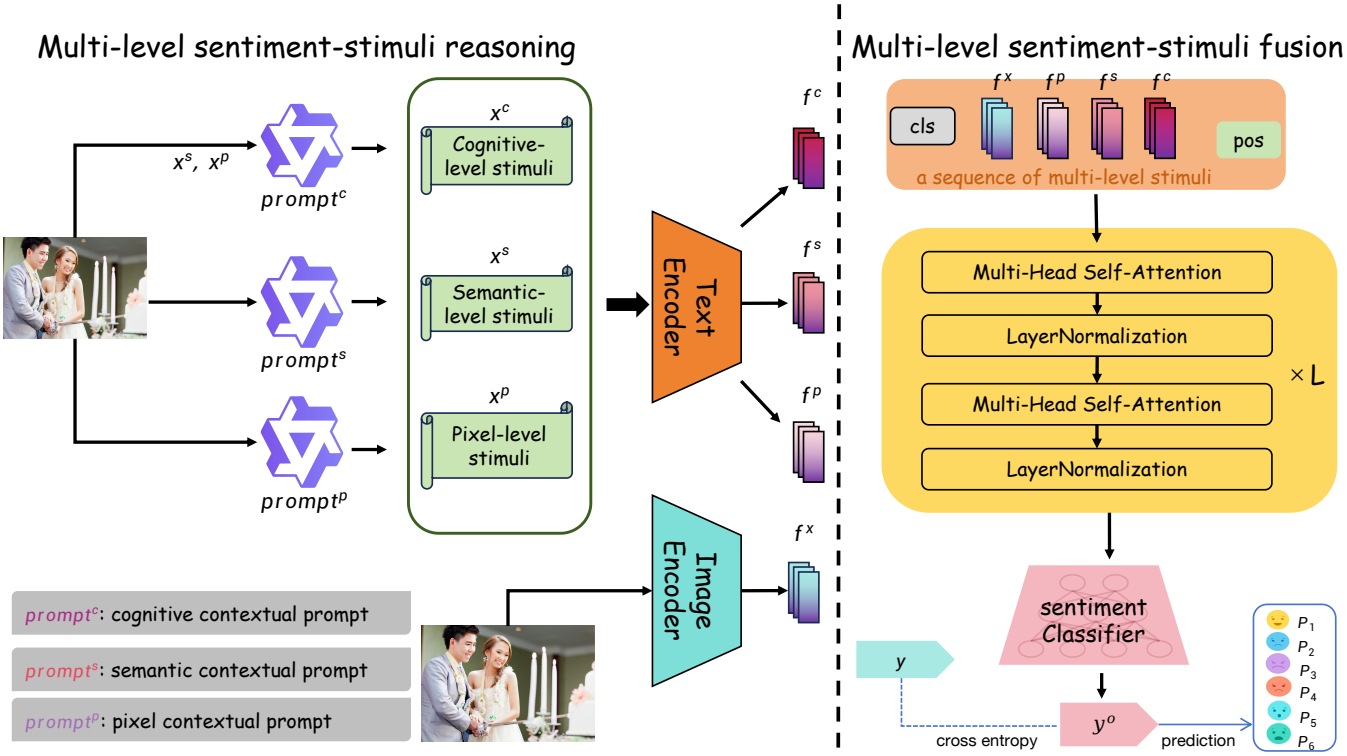

Figure 2: The overall framework of MCRStimuli, which consists of two main stages: (1) multi-level sentiment-stimuli reasoning; (2) multi-level sentiment-stimuli fusion. (see text for details).

Toward this end, we propose a two-stage pipeline. First, we design a self-generation pipeline for comprehensively reasoning the multi-level sentiment-stimuli by employing a VLM. Second, for exploiting all these stimuli for better visual emotion understanding, a transformer-based fusion strategy is introduced for dynamic combination. As a result, visual emotion can be precisely inferred with all the decoded stimuli, which also represent the evidence for the predicted result. The overall framework is shown in Figure 2.

3.1 Multi-level Sentiment Stimuli Reasoning

In this work, we incorporate visual emotional priors for multi-level sentiment-stimuli reasoning. Based on [Machajdik and Hanbury, 2010; Yang *et al.*, 2023], the visual emotion priors include: pixel-level stimuli, semantic-level stimuli, and cognitive-level stimuli. We consider using VLM to reason these sentiment-stimuli based on different contexts, and the corresponding outputs without irrelevant noise are more disentangled for visual emotion understanding. To this end, we design a novel pipeline for automatically reasoning multi-level sentiment stimuli, which is exemplified in Figure 3.

Pixel-level stimuli: capture the general color attributes, which first evoke the human emotions and include the information about colorfulness, brightness, and contrast. First, we enforce the VLM focusing on the global color and lighting information for *pixel-level* stimuli (x^p) generation, which is expressed as follows:

$$x^p = \texttt{VLM}(x, prompt^p), \tag{1}$$

$prompt^p$ is the pixel-level contextual prompt.

Semantic-level stimuli: focus on the key elements within the image, namely human, scenes, and objects, since psychological studies have shown that visual emotions are evoked by the interactions among these semantic cues. Therefore, instead of using the general image captions which may contain non-sentiment information, we make the VLM focus on the corresponding emotional cues. Concretely, the emotional cues of humans include facial expression, body posture, gesture, and social interaction; the emotional cues of objects are summarized by object state; the emotional cues of scenes include density, enclosure, and condition. The semantic-level stimuli can be expressed as follows:

$$x^s = \texttt{VLM}(x, prompt^s), \tag{2}$$

Cognitive-level stimuli: reflect the emotional reasoning process, which consists of the emotion of the image and the reasoning data about why this emotion is evoked. Therefore, our approach extends beyond the objective content, prompting the model to make predictions based on in-depth reasoning cues. Specifically, the VLM is conditioned on the image and the previous objective stimuli during reasoning:

$$x^c = \texttt{VLM}(x, prompt^c, x^p, x^s), \tag{3}$$

3.2 Multi-level Sentiment Stimuli Fusion

After acquiring the sentiment stimuli as detailed in 3.1, our goal can be reformulated to inferring the visual emotion label y with the image x and its corresponding sentiment stimuli

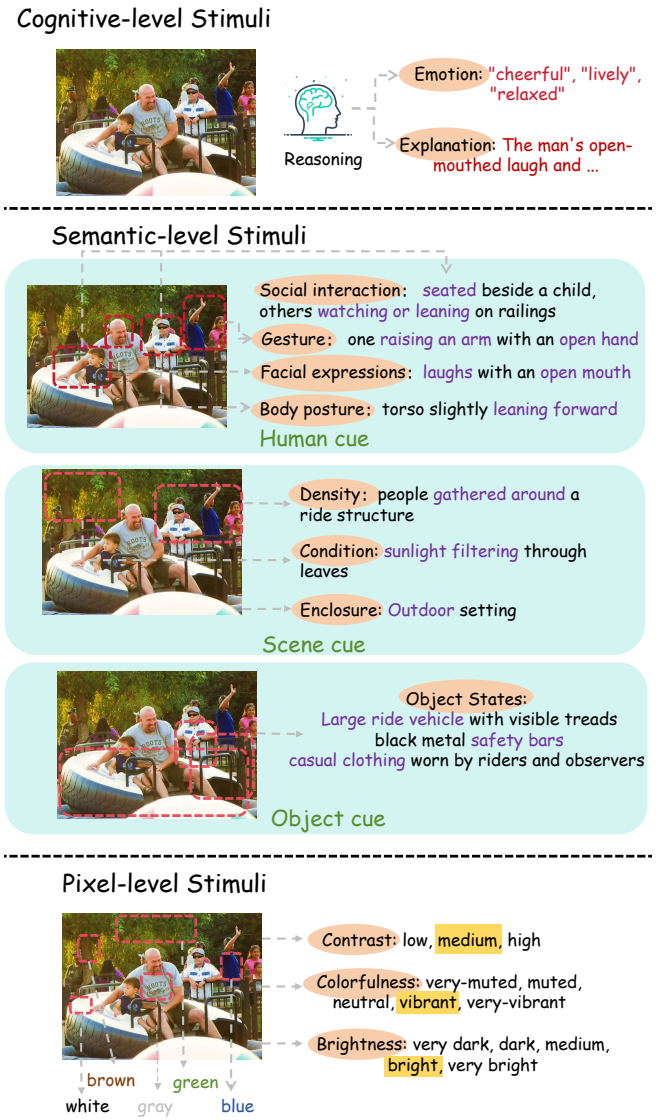

Figure 3: The sample of our multi-level sentiment-stimuli.

$x^{p,s,c}$. To this end, we propose a transformer-based fusion strategy for enhancement, which can dynamically select the optimal information for precise emotion prediction.

Specifically, we first use a pre-trained image encoder and text encoder to extract the features of x and $x^{p,s,c}$, which are expressed as follows:

$$f^x = E_{img}(x),\ f^p = E_{txt}(x^p), \tag{4}$$
$$f^s = E_{txt}(x^s),\ f^c = E_{txt}(x^c), \tag{5}$$

where we use CLIP ViT E_{img} and RoBERTa E_{txt} for semantic visual-linguistic embedding extraction due to their impressive performance.

Then, we obtain a sequence of features $[f^x, f^p, f^s, f^c]$ which represent the sentiment-stimuli of x from different perspectives. Before feeding them to a transformer encoder, a learnable class token $\texttt{cls}$ is appended to the sequence. The input of the transformer can be represented as follows:

$$f_0 = [\texttt{cls}, f^x, f^p, f^s, f^c] + \texttt{pos}, \tag{6}$$

where $\texttt{pos}$ denotes the temporal positions which avoid permutation-invariance in the subsequent self-attention operations, subscript 0 indicates the 0th layer. f_0 is then fed to a fusion encoder with L transformer layers ($L = 2$ in our experiment), where each layer is composed of Multi-Head Self-Attention ($\texttt{MHSA}$), Layer Normalization ($\texttt{LN}$), and Feed Forward Network ($\texttt{FFN}$). The output of lth layer f_l is:

$$u_{l-1} = \texttt{MHSA}(\texttt{LN}(f_{l-1})) + f_{l-1}, \tag{7}$$
$$f_l = \texttt{FFN}(\texttt{LN}(u_{l-1})) + u_{l-1}. \tag{8}$$

The output at the final layer f_L has the same dimension as that of f_0, and we use the first $\texttt{cls}$ token feature for final prediction, which contains all the information of sentiment-stimuli:

$$y^o = \text{softmax}(\mathbf{W} f_l[\texttt{cls}] + \mathbf{b}). \tag{9}$$

3.3 Model Optimization

In practice, all the features in f_0 of Eq. (6) can be considered as the image-supplementary feature for emotional prediction. Besides, to avoid the dominant feature steering the learning process, which leads to insufficient training of other stimulus features, we introduce an additional classification head for each feature and optimize the corresponding prediction loss to retain the emotion-predictive capability. The corresponding prediction logits are:

$$y^i = \text{softmax}(\mathbf{W}^i f^i + \mathbf{b}^i), \tag{10}$$

where $i \in \{c, p, s\}$.

By integrating all these sentiment-stimuli, the overall optimization process can be expressed as follows:

$$\mathcal{L} = -\sum_{i \in \{c,p,s\}} [y \log(y^i)] - y \log y^o, \tag{11}$$

where y is the ground truth. We train the overall framework in an end-to-end manner. During inference, we use the y^o as the prediction result.

4 Experiments

4.1 Datasets

Following previous works [Deng *et al.*, 2023; Zhou *et al.*, 2025b], we evaluate MCRStimuli on five widely-adopted visual emotion recognition datasets: EmoSet [Yang *et al.*, 2023], FI [You *et al.*, 2015], EmotionROI [Peng *et al.*, 2016], Twitter I [Borth *et al.*, 2013], and Twitter II [You *et al.*, 2015].

EmoSet is the largest dataset, which contains 118,102 images and maintains a balanced distribution across 8-class emotions based on the Mikels model. In terms of reliability, each image is annotated by 10 human annotators.

FI is collected from Flickr and Instagram by searching eight emotion categories from the Mikels model as keywords. It contains 22,683 images, and their eight emotion categories can be split into positive (amusement, awe, contentment, excitement) and negative (anger, disgust, fear, sadness). The

Table 1: Quantitative comparison of MCRStimuli and competing methods on different visual emotion datasets. The best results are denoted with **bold**. The reported values for the competing methods are taken directly from the original publication.

Category	Method	EmoSet		FI		EmotionROI		Twitter I	Twitter II
Number of classes		8	2	8	2	6	2	2	2
Generic VEA	ResNet101 [He *et al.*, 2016]	0.7487	0.9384	0.6616	0.7576	0.5160	0.7392	0.7813	0.7823
	WSCNet [She *et al.*, 2019]	0.7632	0.9416	0.7007	–	0.5825	–	0.8425	0.8135
	SOLVER [Yang *et al.*, 2021a]	–	–	0.7233	–	0.6212	–	0.8543	0.8319
	Stimuli-aware [Yang *et al.*, 2021b]	0.7840	0.9458	0.7242	–	0.6162	0.8228	0.7983	
	DeepSentibank [Chen *et al.*, 2014]	–	–	0.5154	0.6154	0.4253	0.7011	0.7125	0.7023
	MDAN [Xu *et al.*, 2022]	0.7575	0.9371	0.7641	0.9108	0.6166	0.8462	–	–
VLM-assisted VEA	CLIP [Radford *et al.*, 2021]	0.6976	0.9074	0.7784	0.8906	0.6565	0.8773	0.8710	0.8145
	CoCoOp [Zhou *et al.*, 2022]	0.8031	0.9333	0.7746	0.9124	0.7109	0.8910	0.9163	0.8294
	SimEmotion [Deng *et al.*, 2023]	0.7906	0.9428	0.8033	0.9542	0.7054	0.9040	0.8976	0.8421
	PT-DPC [Deng *et al.*, 2024]	0.7713	0.9246	0.7807	0.9389	0.6970	0.8855	0.9094	0.8250
	MASANet [Cen *et al.*, 2024]	–	–	0.7916	0.9507	0.7323	0.9175	0.9216	0.8870
	MVP [Xu *et al.*, 2024]	0.8192	0.9644	0.8276	0.9698	0.7189	0.9255	0.9203	0.8821
	EmoViT [Xie *et al.*, 2024]	0.8336	–	0.6809	–	0.5387	–	–	–
	EmoVLM-KD [Lee *et al.*, 2025]	0.7983	–	0.7951	–	–	–	–	–
	ESCNet [Zhou *et al.*, 2025b]	–	–	0.8086	0.9571	0.7399	0.9186	0.9246	0.8926
	MCRStimuli (ours)	**0.8350**	**0.9705**	**0.8290**	**0.9701**	**0.7727**	**0.9644**	**0.9488**	**0.8992**

Table 2: Ablation studies of different sentiment-stimuli in our framework. The best results are in **bold**.

Pixel	Semantic	Cognitive	EmotionROI		Twitter I		Twitter II		FI		EmoSet	
			Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
✓			0.7424	0.7410	0.9370	0.9330	0.8739	0.8230	0.8016	0.7960	0.8213	0.8300
	✓		0.7526	0.7540	0.9349	0.9420	0.8892	0.8520	0.8199	0.8050	0.8201	0.8280
		✓	0.7626	0.7630	0.9409	0.9370	0.8982	0.8610	0.8231	0.8130	0.8292	0.8390
✓	✓		0.7586	0.7640	0.9349	0.9320	0.8922	0.8530	0.8199	0.8050	0.8201	0.8280
	✓	✓	0.7677	0.7620	0.9450	0.9420	0.8892	0.8480	0.8231	0.8130	0.8314	0.8410
✓	✓	✓	**0.7727**	**0.7710**	**0.9488**	**0.9480**	**0.8992**	**0.8630**	**0.8290**	**0.8150**	**0.8350**	**0.8420**

label of each image is determined by the consensus of at least three out of Amazon Mechanical Turk (AMT) workers.

EmotionROI is retrieved from Flickr by querying six Ekman emotion words and their synonyms (i.e., anger, disgust, fear, joy, sad, surprise). Besides, joy and surprise are positive labels, and anger, disgust, fear, sad are negative labels. The label distribution of EmotionROI is class-balanced (330 images per emotion), resulting in 1,980 images in total.

Twitter I and **Twitter II** are two small visual emotional benchmarks collected from the Twitter website. Both are labeled with positive and negative. Twitter I contains 1,269 images with 5 AMT annotators per image, while Twitter II contains 603 images with 3 AMT annotators per image.

4.2 Implementation Details

For fair comparisons with previous works [Deng *et al.*, 2023; Xu *et al.*, 2024; Lee *et al.*, 2025], the EmoSet and FI are split into training set (80%), validation set (5%), and testing set (15%). For the remaining smaller datasets, we randomly split them into a training set (80%) and a testing set (20%). We conduct the experiments 5 times with random seeds and report the average accuracy for all the datasets. In addition, we standardize the labels from both EmoSet and FI into positive and negative categories and evaluate the model's binary classification accuracy accordingly.

Following the standard protocols [Deng *et al.*, 2024; Zhou *et al.*, 2025b], we use the pre-trained CLIP ViT-B/32 [Radford *et al.*, 2021] as the image encoder and RoBERTa [Liu *et al.*, 2019] as the text encoder. The transformer encoder for fusion is a lightweight module with two self-attention layers. We use AdamW [Zhou *et al.*, 2024] optimizer during training. Without specification, the learning rates for the image encoder and the text encoder are set to 1×10^{-6}, and the learning rate for the fusion transformer is 1×10^{-4}. For small datasets, Twitter I and Twitter II, we only train the fusion transformer to avoid overfitting. We set the batch size to be 32, the weight decay to be 1×10^{-2}, and the number of epochs to be 50. All the experiments are conducted on a NVIDIA RTX 3090 GPU with 24GB of memory.

4.3 Baseline Models

We compare our MCRStimuli with recent representative and state-of-the-art methods: ResNet [He *et al.*, 2016], WSCNet [She *et al.*, 2019], Stimuli-aware [Yang *et al.*,

2021b], and SOLVER [Yang *et al.*, 2021a] are vision-based models that only rely on extracting generic or sentiment-related visual features for prediction. DeepSentibank [Chen *et al.*, 2014] introduced external ANP-based concepts as auxiliary information for supervision. MDAN [Xu *et al.*, 2022] leveraged structured affect knowledge defined by Parrott for enhancement. Recently, SimEmotion [Deng *et al.*, 2023], PT-DPC [Deng *et al.*, 2024], and MVP [Xu *et al.*, 2024] proposed to leverage emotion-related text prompts to realize language supervision with the pre-trained CLIP model, which shared a similar idea to the conditional prompt learning (CoCoOp) [Zhou *et al.*, 2022]. EmoViT [Xie *et al.*, 2024] adopted instruction tuning on BLIP [Li *et al.*, 2023] with the self-generated emotion visual instruction data, EmoVLM-KD [Lee *et al.*, 2025] augmented EmoViT with a lightweight module distilled from conventional vision models. MASANet [Cen *et al.*, 2024] and ESCNet [Zhou *et al.*, 2025b] directly generated the emotional descriptions for improvement. We categorize these methods into generic VEA and VLM-assisted VEA, which are shown in Table 1.

4.4 Quantitative Comparison

In this section, we perform quantitative comparisons between MCRStimuli and competing methods; the corresponding results on different visual emotion datasets are reported in Table 1. It should be noted that generic VEA methods mainly rely on the visual features, while VLM-assisted VEA leverages the pre-trained VLM for providing additional signals. According to Table 1, there are following observations:

- The VLM-assisted methods consistently outperform the generic methods, which implies that there exists a significant affective gap between the visual feature and the emotional concepts; thus, leveraging VLM to bridge the gap in our model is important for delivering reliable results.
- Some methods (CoCoOp, SimEmotion, PT-DPC) regarded the visual emotion analysis as a traditional classification task, and proposed to inject visual- or class-relevant information into the text prompts for better guidance. The corresponding results are not significantly improved. Therefore, integrating both visual and text information in our model can lead to more enhanced results, which is also demonstrated in [Cen *et al.*, 2024].
- Our MCRStimuli outperforms all the related competing methods in terms of accuracy across 5 visual emotion analysis tasks. In particular, MASANet proposed to generate multi-aspect descriptions for fusion, and our MCRStimuli significantly surpasses it on the FI and EmotionROI datasets by around 3 to 4 percentage points in terms of accuracy. Additionally, the recent model ESCNet demonstrated improved performance on Twitter I and Twitter II by generating semantic-level descriptions. While MCRStimuli obtains better performance on both datasets, which verifies the superiority of our model by reasoning the multi-level sentiment-stimuli.

Table 3: Ablation studies of different aspects of semantic-level stimuli in our framework. The best results are in bold.

H	S	O	Twitter I		Twitter II		EmotionROI	
			Acc	F1	Acc	F1	Acc	F1
✓			0.9249	0.9230	0.8808	0.8160	0.7426	0.7400
	✓		0.9331	0.9290	0.8852	0.8380	0.7327	0.7390
		✓	0.9291	0.9250	0.8792	0.8320	0.7326	0.7300
✓	✓	✓	0.9439	0.9420	0.8892	0.8520	0.7526	0.7540

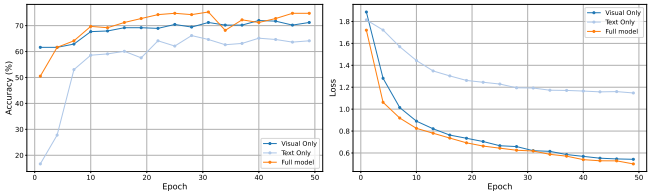

Figure 4: Comparison the performance of visual stimuli and text stimuli on the EmotionROI dataset: (a) Loss and (b) Accuracy (%) w.r.t. Epoch.

4.5 Ablation Study

In this section, we conduct several ablation studies to validate the effectiveness of the proposed modules. The qualitative and quantitative results are presented in Table 2, Table 3, Table 4 and Figure 4, respectively.

(a) Multi-level sentiment-stimuli. We first analyze the effects of multi-level sentiment-stimuli generated by VLM with different contextual prompts. As shown in Table 2, removing any level stimuli leads to a clear performance drop, and these multi-level sentiment-stimuli have different effects on various visual emotion benchmarks. In addition, the pixel-level stimuli yield limited improvements, and semantic-level and cognitive-level stimuli often produce better results. In practice, the cognitive-level stimuli are already competitive, suggesting that reasoning the affective signals with the surficial stimuli is important. Overall, including multi-level stimuli delivers the best performance, which verifies the motivation of MCRStimuli.

(b) Multi-aspect of semantic-level stimuli. We further study the contributions of Human (H), Scene (S), and Object (O) in the semantic-level stimuli, which are widely considered in previous works [Yang *et al.*, 2021b]. As shown in Table 3, the scene plays a more important role in emotion recognition for the Twitter datasets, whereas the effect of human aspect is more significant in the EmotionROI dataset. And as expected, considering all these semantic aspects often leads to the best results.

(c) Fusion of visual and text stimuli. Instead of directly using the VLM-generated stimuli for visual emotion analysis, our MCRStimuli fused both visual and text stimuli for emotion prediction. Table 4 shows that our full model outperforms both visual-only and text-only variants, which indicates that both information are complementary to each other. In addition, as shown in Figure 4, our full model not only achieves the best results, but also converges faster than the model based on visual or reasoned stimuli.

GroundTruth	Different Stimuli	Prediction
anger	Pixel-level stimuli: The overall tone is medium and vibrant with brown, red, teal, black, yellow colors.	Lable: excitement Confidence:0.4885
	Semantic-level stimuli: Specifically, This image shows Two men stand face-to-face in a ring, one in a long coat, the other in wrestling gear, with spectators watching closely. The scene features Indoor arena with red curtains and bright overhead lights, audience seated close to the ring, focused on the confrontation. Notable objects include Wrestling ring ropes, black mat surface, and photographer's camera visible near the edge of the ring.	Lable: anger Confidence:0.8840
amusement	Semantic-level stimuli: Notable objects include layered translucent orange-pink paper fragments form a voluminous, textured mass suspended in air.	Lable: disgust Confidence:0.7292
	Cognitive-level stimuli: Specifically, The layered translucent orange-pink fragments create a gentle, voluminous texture under bright lighting, evoking warmth and softness, while their floating arrangement lends a dreamy, ethereal quality. These cues indicate a warm, soft, and dreamy atmosphere.	Lable: amusement Confidence:0.9207
fear	Only Scene aspect: The scene features A dimly lit bathroom with a person lying on the floor near a bathtub, walls and fixtures intact but shadows casting over the space. Specifically, the low brightness and enclosed spatial composition enhance a sense of stillness and unease. These cues indicate a gloomy, oppressive, and still atmosphere.	Lable: anger Confidence:0.5158
	Human and Scene aspects: This image shows A person lies on the bathroom floor with legs raised and arms extended, wearing a loose shirt, head turned away from view, body partially obscured by shadow. Specifically,The person's prone, partially obscured posture in a confined space evokes unease, while the muted color palette enhances the gloomy mood without direct emotional cues. These cues indicate a gloomy, oppressive, and unsettling atmosphere.	Lable: fear Confidence:0.5558

Figure 5: Case analysis of our model with different sentiment stimuli. The correct and incorrect predictions are denoted with green and red, respectively.

Table 4: Ablation studies of the visual and text stimuli in our framework. The best results are in bold.

Dataset	Vision-only	Text-only	Full model
EmotionROI	0.7197	0.7071	**0.7727**
Twitter1	0.9252	0.9055	**0.9488**
Twitter2	0.8430	0.8235	**0.8992**
FI	0.7890	0.7533	**0.8290**
Emoset	0.8210	0.8043	**0.8350**

Table 5: Held-out performance comparison on visual emotion datasets.

Method	**EmoSet**	**FI**	**EmotionROI**
LLaVA	0.4403	0.5604	0.4646
BLIP2	0.4679	0.5772	0.5051
InstructBLIP	0.4220	0.3797	0.4613
EmoViT	0.8336	0.6809	0.5387
MCRStimuli (Ours)	**0.8350**	**0.6955**	**0.6061**

4.6 Model Analysis

In this section, we further consider held-out evaluation to study the generalization of our model. Following EmoViT [Xie *et al.*, 2024], we train our model on EmoSet due to its comprehensive inclusion of emotion attributes. Then, the model will be tested on the other datasets. Table 5 provides the results of different VLM models. These VLMs excel in visual understanding but lack the ability for emotion analysis without tuning, directly inquiring the emotions of images often leads to incorrect responses. EmoViT improves the emotional understanding by instruction tuning with generated emotional data, and shows good generalization performance. In contrast, our MCRStimuli achieves better performance compared with EmoViT, especially on the EmotionROI dataset (53.87% vs. 60.61%). These results confirm the generalization of our model by reasoning multi-level sentiment-stimuli.

4.7 Case Analysis

In this section, we present several cases for analyzing the behaviors of our model with different sentiment stimuli. For the first case, the simple pixel-level stimuli (the color and contrast signals) lead to incorrect prediction, and the semantic-level stimuli, which illustrate what happened in the image, can yield the correct emotion. In addition, for more vague samples (the last two cases), reasoning the emotion beyond the image with the pixel- and semantic-level stimuli plays the key role, and the superficial sentiment cues often result in emotional misunderstanding. In the last case, we also demonstrate that the human aspect in the semantic-level stimuli is important for the precise prediction.

5 Conclusion

This paper proposes MCRStimuli, a novel framework for visual emotion analysis that reasons multi-level sentiment stimuli. Departing from existing methods that rely solely on visual features or descriptions, we first categorize sentiment stimuli into three distinct levels: pixel, semantic, and cognitive. Our approach then integrates these stimuli, as reasoned by a Vision-Language Model (VLM), leading to significant performance gains. Comprehensive experiments validate the framework's effectiveness and generalization across multiple visual emotion benchmarks, offering valuable insights for advancing community research in this direction.

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